# 2. Project Proposal: E-Commerce Product Recommendation System using Python

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## 1. Introduction / Background

In today's digital world, e-commerce platforms have become an essential part of online shopping. With a vast number of products available, users often face difficulty in finding relevant items according to their preferences. Recommendation systems play a vital role in improving user experience by suggesting products based on user behavior and interests.

This project aims to develop an E-Commerce Product Recommendation System using Python that recommends products to users based on their past interactions and preferences using machine learning techniques.

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## 2. Problem Statement

The problem is to design an intelligent system that can analyze user behavior and recommend relevant products, helping users discover items of interest and improving their shopping experience.

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## 3. Objectives

- To build a recommendation system for e-commerce products.
- To analyze user behavior such as browsing history or purchase data.
- To implement a recommendation technique such as content-based or collaborative filtering.
- To provide personalized product suggestions to users.
- To develop a simple interface for user interaction.

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## 4. Methodology

- **Data Source** – Use a publicly available dataset such as Amazon Product Dataset or any e-commerce dataset from Kaggle.
- **Data Preprocessing** – Clean and organize data, including user interactions, product details, and ratings.

- **Feature Extraction** – Extract relevant features such as product categories, user preferences, and ratings.
- **Recommendation Model** – Implement a recommendation algorithm such as content-based filtering or collaborative filtering.
- **Evaluation** – Evaluate system performance using metrics such as accuracy or user satisfaction.
- **Prediction System** – Recommend products to users based on their preferences and past interactions.

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## 5. Expected Outcomes

- A working recommendation system for e-commerce products.
- Personalized product suggestions for users.
- Improved understanding of recommendation algorithms.
- A simple and user-friendly interface.

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## 6. Tools and Technologies

- **Programming Language:** Python
- **Libraries:** Pandas, NumPy, Scikit-learn
- **Techniques:** Content-Based Filtering / Collaborative Filtering
- **Dataset:** Amazon Product Dataset (Kaggle)
- **Environment:** Jupyter Notebook or Google Colab
- **Deployment:** Streamlit or Flask

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## 7. Timeline

- **Week 1:** Dataset collection and preprocessing
- **Week 2:** Feature extraction and model development
- **Week 3:** Recommendation system implementation
- **Week 4:** Testing, deployment, and report preparation

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## 8. References


- Kaggle Dataset – https://www.kaggle.com/
- Scikit-learn Documentation – https://scikit-learn.org/
- Pandas Documentation – https://pandas.pydata.org/
- Streamlit Documentation – https://streamlit.io/